

- 5 (a) Define electric field.

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..... [2]

- (b) Fig. 5.1 shows two parallel conducting plates that are in a vacuum. The plates are separated by a distance of 6.7 cm and have a potential difference (p.d.) of 430 V between them.

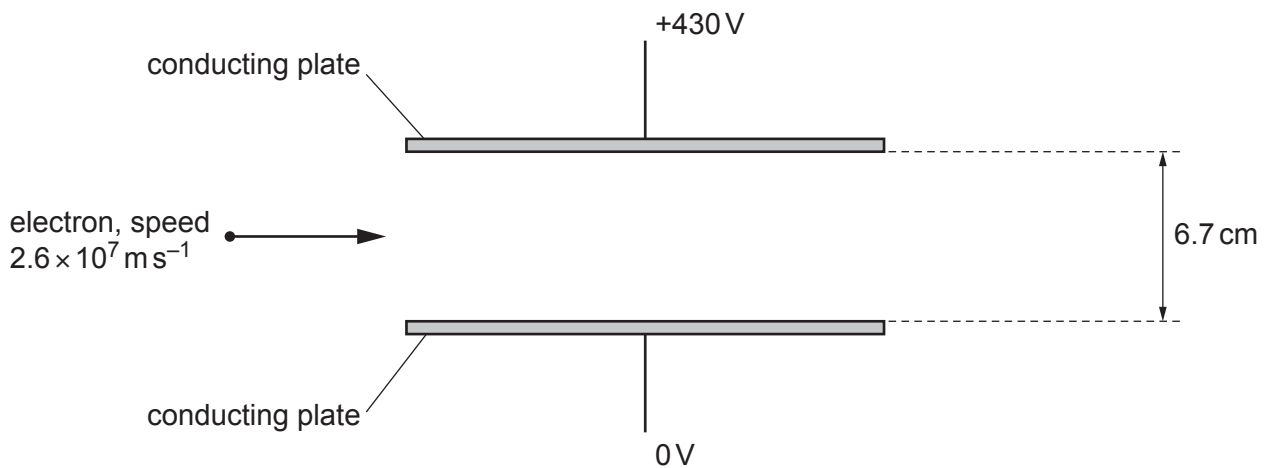


Fig. 5.1

- (i) On Fig. 5.1, draw four field lines to represent the electric field between the plates. [2]
- (ii) Determine the strength E of the electric field between the plates.

$$E = \dots\dots\dots \text{NC}^{-1} \text{ [2]}$$

- (iii) An electron travels at a speed of $2.6 \times 10^7 \text{ ms}^{-1}$ towards the region between the plates, as shown in Fig. 5.1.

On Fig. 5.1, draw the path of the electron as it moves between and beyond the plates. [2]

- (c) A uniform magnetic field is now applied in the region of the electric field in Fig. 5.1, so that the electron in (b)(iii) travels undeviated through the region.

- (i) Determine the direction of the uniform magnetic field.

..... [1]

- (ii) Explain, with reference to the forces exerted by the two fields on the electron, why the path of the electron is undeviated.

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..... [2]

- (iii) Determine the flux density B of the uniform magnetic field. Give a unit with your answer.

$B =$ unit [2]

[Total: 13]